Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

1. Student Information

Field	Value
Student name (printed):	**LÊ PHẠM KIỀU DUYÊN**
Student ID:	**23127184**
Class / Cohort:	**CSC13003 – Software Testing**
Assignment ID (e.g., HW#00, HW#02):	**HW#01**
Assignment date:	May 19, 2026
AI tool(s) used:	**Gemini, Claude**
AI tool(s) used:	**[x] Yes** [ ] No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Artifact #1 **Tool: Gemini** **Time: 2026-06-01 15:01:24** **Prompt: "Act as an expert software QA/QC engineer. Please help me find 20 real-world software defects publicized between 2022 and 2026 for my Software Testing assignment.** **The output must strictly satisfy the following criteria:** **1. Provide exactly 20 distinct software defects.** **2. At least 5 of these defects must be directly related to AI/LLM systems (such as hallucinations, prompt injection, bias, or safety guardrail failures).** **3. For EACH defect, you must provide the following information using clear headings or bullet points:** **- Source Link: (A real, verifiable URL or reference)** **- Description: (What the bug was and how it occurred)** **- Severity: (Low / Medium / High / Critical)** **- Consequences: (The direct impact or damage caused)** **- Solution: (How the issue was fixed, patched, or mitigated)** **Please format the response clearly so it is easy to read."**	**Link markdown:** [artifact01]()	INCOMPLETE	At first glance, the defect generated by the AI looked realistic and convincing. However, after checking the information, I found that it was completely made up. The National Vulnerability Database (NVD) had no record of CVE-2025-99812, while "CyberDyne Systems" and "QuantumOS" turned out to be fictional names. The CISA link provided by the AI was also invalid and returned a 404 error. This was a clear example of the AI creating information that sounded believable but was not real. Because the defect could not be verified, it did not meet the requirement of using real-world examples. It also went against the ISTQB principle of working with accurate and reliable information during testing. As a result, I removed the defect from the report and replaced it with a verified real-world case.	The fraudulent CyberDyne entry has been completely rejected and purged from the final report. I have replaced it with a verified, real-world supply chain attack: the **Mini Shai-Hulud npm Worm Attack (May 2026)**, backed by legitimate Microsoft Security and Akamai Research documentation. The fully corrected defect details—including the authentic description, severity, consequences, and solutions—have been properly formatted and documented inside the repository file: **artifact01_fix.md** (please click the file link in the repository to view). Link markdown: [artiface01_fixed]()
Artifact #2 **Tool: Gemini** **Time: 2026-06-01 16:00:05**	**Link markdown:** [artifact02]()	INCOMPLETE	- **For Error 01:** The AI incorrectly assigned "Source Code Debugging and Patching" to the QC Tester role. According to standard software engineering and	Mindmap after fix: - **Link PNG:** [mindmap_fixed]()

Prompt: “Act as an ISTQB-certified QA/QC expert. Please generate a detailed QA/QC mindmap focusing on QA/QC roles, standard testing processes, and required skills in the current 2026 AI-augmented job market. **Please format the output strictly using mermaid code so that I can easily visualize it or convert it into diagram.”**	**Link PNG:** mindmap		ISTQB principles, debugging (identifying the root cause of a failure) and patching (fixing the source code) are the definitive responsibilities of a Developer. The QC Tester's role is strictly bounded to designing tests, executing them, and logging/tracking defects, not modifying the codebase. - **For Error 02:** The AI violated the chronological logic of the standard testing process by placing "Test Execution and Logging" before "Test Analysis and Design". As mandated by the ISTQB Foundation Level Syllabus, a testing team must analyze requirements and design test cases first to establish solid test conditions. Execution cannot logically or systematically happen before these foundational phases are complete. - **For Error 03:** The AI introduced a dangerous deployment anti-pattern by proposing "Autonomous AI Sign-off" and "Zero Human Verification Production Releases" for the 2026 landscape. This directly breaches core quality governance and risk management principles. Modern software testing frameworks and responsible AI safety guidelines strictly dictate a "Human-in-the-loop" approach to independently verify AI-generated artifacts before production deployment.	
Artifact #3 **Tool: Claude** **Time: 2026-06-01 15:03:57** **Prompt: “Đóng vai một nhà kiểm thử phần mềm chuyên nghiệp. Hãy giúp mình thiết kế 15 test case để test chiếc quạt điện tử trong nhà. Mô tả quạt: Senko, model TR1683 (Mô tả sản phẩm…)”**	**Link** **artifact03**	VALID	The 15 baseline test cases from the AI are good for basic functions like power, speed, and the timer. However, the suite is incomplete because the AI missed complex hardware risks. To fix this, I added 3 necessary edge cases: - **Power Brownout (TC-16):** Checks if the fan keeps its memory and settings during a sudden voltage drop. - **IR Signal Occlusion (TC-17):** Verifies if the fan recovers smoothly if the remote signal is blocked mid-command. - **Timer Expiry Race Condition (TC-18):** Tests if the fan shuts down cleanly when a mode change happens at the exact same second the timer ends.	I reviewed and accepted the 15 baseline functional test cases as-is. To satisfy the advanced edge-case requirement of the assignment and ensure robust physical device testing, I have manually appended 3 advanced edge cases (**TC-16: Power Brownout**, **TC-17: IR Signal Occlusion**, and **TC-18: Timer Expiry Race Condition**) into the matrix. The fully expanded 18-test case suite has been updated with real physical execution verdicts and is documented completely inside the project artifact files.

4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	3	100%
VALID (correct, accepted as-is)	1	33,3%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	2	66,7%

5. Conclusion — When should AI be used (or not)?

Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

AI is highly effective as an initial assistant, proving excellent at sketching out basic ideas and quickly drafting mind maps or simple test scenarios. However, a clear pattern of limitations emerged: the tool is overly optimistic, focusing mostly on ideal situations while getting easily confused by wording. It failed to separate standard system glitches from unique logic errors, and it completely missed real-world physical problems, such as simultaneous commands or sudden interruptions during hardware tests.

Therefore, AI should only be used as a starting tool to create rough drafts. Human oversight remains absolutely mandatory to test for real-world unpredictability and catch the complex errors that AI overlooks.

6. Mandatory Disclosure (paste verbatim)

> *"[Test cases / script / dataset / report] was initially generated by [AI tool name]; I reviewed and modified [section X], added [edge cases Y, Z]; [section W] was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category."*

The test cases and QA/QC mindmap were initially generated by Gemini; I reviewed and modified the mindmap to meet ISTQB standards and the test case matrix to eliminate invalid or unsafe routines, added edge cases regarding physical signal conflicts and boot-time button spamming; Requirement 1 (QA/QC Job Market 2026+) and the AI Critique section were written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

Student name (printed):	**LÊ PHẠM KIỀU DUYÊN**
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Class / Cohort:	23KTPM2
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Lâm Quang Vũ/ Trương Phước Lộc/ Hồ Tuấn Thanh
Date:	May 30, 2026
Signature:	

References

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.